DR. SIDDHARTH MUKHERJEE

Senior Research Scholar & Distributed Systems Specialist

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Portfolio

PROFESSIONAL SUMMARY

Doctoral researcher with 5+ years of extensive research in consensus algorithms, Byzantine fault tolerance, and verifiable computing. Author of 6 peer-reviewed conference publications with substantial industrial engineering contributions across distributed storage engines.

EDUCATION

Indian Institute of Technology Delhi 2021 – 2025
Ph.D. in Computer Science & Engineering in Distributed Systems & Cryptography **CGPA / Score: 9.45 / 10.0**

Indian Institute of Science Bangalore 2019 – 2021
Master of Technology in Computer Science in Computer Systems **CGPA / Score: 9.60 / 10.0**

Jadavpur University 2015 – 2019
Bachelor of Engineering in Computer Science in Computer Science **CGPA / Score: 8.88 / 10.0**

SKILLS

Technical: Rust · C++ · Go · Distributed Consensus (Raft/Paxos) · Zero-Knowledge Proofs · Linux eBPF · RocksDB · Formal Verification (TLA+) · PostgreSQL

Soft Skills: Doctoral Mentorship · Grant Proposal Writing · Research Dissemination · Technical Leadership

PROJECTS

VeriSync: High-Throughput BFT State Machine Principal Researcher

A novel Byzantine fault-tolerant consensus protocol decoupling transaction ordering from execution pipeline.

- Formally verified safety and liveness invariants using TLA+ specification checker.
- Achieved 140,000 transactions/second with sub-second finality across a 64-node geo-distributed WAN testbed.
- Optimized network serialization using zero-copy byte buffers cutting CPU cycles per block by 32%.
- Published architecture paper in ACM Symposium on Operating Systems Principles (SOSP 2024).

Technologies: Rust, Tokio, RocksDB, BLS Signatures

Live Demo · Source Code

KernelFlow: eBPF Network Observability Daemon Systems Programmer

Kernel-space packet tracking agent monitoring TCP socket lifecycle and socket buffer queue delays.

- Captured TCP retransmits and window stalls directly in kernel space with negligible (< 0.8%) overhead.
- Exported OpenTelemetry flow traces to Grafana tempo for microservice latency bottleneck analysis.

Technologies: C, eBPF, Go, Prometheus

Source Code

PebbleStore: Append-Only Write-Optimized Log Core Contributor

Storage engine optimized for NVMe SSD drives taking advantage of Linux io_uring asynchronous system calls.

- Surpassed standard synchronous append benchmarks by 3.8x under random multi-threaded workloads.
- Implemented lock-free ring buffer for write request batching and concurrent epoch flushes.

Technologies: C++, Direct I/O, io_uring

Source Code

CertiCrypt: Formally Verified Cryptographic Library — Lead Developer

Constant-time elliptic curve arithmetic implementation formally verified against timing side-channel leaks.

- Machine-checked proof in Coq guaranteeing absence of integer overflow and timing variance.

Technologies: OCaml, Coq, C

Source Code

AegisDB: Hardware-Enclave Encrypted Database — Co-Investigator

Confidential database engine executing relational queries inside hardware-protected memory enclaves.

- Mitigated memory access pattern side channels via oblivious RAM (ORAM) tree shuffling.
- Sustained 4,500 encrypted SQL transactions per second with under 12% enclave context-switch overhead.

Technologies: C++, Intel SGX, SQLite, AES-GCM

Source Code

SimMesh: Peer-to-Peer Epidemic Dissemination Simulator — Research Programmer

Large-scale discrete event simulator modeling probabilistic gossip protocols across 50,000 simulated nodes.

- Analyzed message propagation latency across diverse power-law network graphs.
- Presented findings at ACM SIGMETRICS student poster track.

Technologies: Python, SimPy, NetworkX, NumPy

Source Code

EXPERIENCE & INTERNSHIPS

Research Intern — Systems & Networking Group — Microsoft Research — Jun 2023 – Dec 2023
Redmond, WA (Remote)

Investigated multi-tenant performance isolation in distributed storage engines for Azure cloud infrastructure.

- Designed weighted fair-queuing scheduler preventing noisy neighbors from starving tail latency sensitive storage requests.
- Filed 2 US patents on adaptive token-bucket rate limiting for cloud shared storage volumes.

Software Engineering Intern — Cloud Spanner — Google Cloud Platform — May 2020 – Aug 2020
Hyderabad, India

Worked on Paxos leader election stability under asymmetric network partitioning.

- Simulated network disruption topologies using Jepsen testing framework to validate fault tolerance.
- Improved failover detection latency by 28% without triggering false leader elections.

Head Teaching Assistant & Lab Coordinator — IIT Delhi Department of CSE — Jan 2022 – May 2024
New Delhi, India

Managed 8 subordinate teaching assistants and oversaw lab curriculum for Advanced Operating Systems course.

- Authored 4 complex programming assignments on virtual memory paging and xv6 operating system kernel hacking.

Member of Technical Staff — Core Database Engine — Oracle India Development Center — Jul 2019 – Jul 2021
Bangalore, India

Maintained redo log buffer caching and write-ahead logging (WAL) serialization routines.

- Patched race condition in shared memory lock acquisition under extreme latch contention.
- Received Engineering Excellence spot award for zero customer-reported regressions in database release 21c.

CERTIFICATIONS & COURSES

Certified Information Systems Security Professional (CISSP) · (ISC)² — 2022

ACHIEVEMENTS & HONORS

- Prime Minister Research Fellowship (PMRF) awardee for exceptional doctoral research (2021–2025).
- Best Paper Award at Asia-Pacific Systems Workshop (APSys 2023).
- ACM ICPC World Finalist representing India at Porto, Portugal (2019).

LEADERSHIP & RESPONSIBILITY

General Secretary — Association for Computing Machinery (ACM) Student Chapter — IIT Delhi 2022 – 2024

Organized national technical symposium with 1,200 participants and 18 invited corporate tech talks.

- Spearheaded diversity in computing initiative increasing female conference participation by 35%.

Peer Reviewer — IEEE Transactions on Dependable and Secure Computing 2023 – Present

Reviewed 14 manuscript submissions on consensus protocols and distributed fault models.

RELEVANT COURSEWORK

Advanced Distributed Systems · Formal Methods & Verification · Applied Cryptography · Kernel Architecture & Virtualization · Storage Systems Engineering